

1. Introduction

1.1 Program Overview

The mainframe application suite is a comprehensive set of tools designed to manage and process various business critical processes for an enterprise. The suite consists of multiple interconnected components that work together to provide a robust and efficient solution.

The application suite utilizes COBOL programming language for developing and maintaining the core logic of the system, which includes data processing, reporting, and business rules management. The COBOL programs are compiled into executable code that runs on the z/OS operating system.

JCL (Job Control Language) jobs are used to manage the execution of these COBOL programs. JCL jobs are used to schedule, monitor, and control the execution of COBOL programs, as well as other tasks such as data transfers and reporting.

The application suite also leverages BMS (Business Management System) mapsets to define the mapping between the COBOL program variables and the database records. This enables the system to accurately retrieve and manipulate data from the underlying database.

Assembler modules are used to optimize performance-critical sections of code, where the z/OS operating system provides a low-level API for direct memory manipulation. These optimized modules are then integrated with the rest of the COBOL codebase.

The application suite is designed to be highly configurable and extensible, allowing users to tailor the system to meet their specific business needs. The integration of multiple technologies and frameworks enables the development of complex systems that can handle a wide range of tasks, from data processing and reporting to user interfaces and business rules management.

1.2 Objectives

The primary objectives of this application suite are:

- To provide a scalable and flexible solution for managing business critical processes.
- To enable real-time data processing and reporting capabilities.
- To ensure data integrity and security through robust data validation and verification mechanisms.
- To support multi-user and multi-process environments, with efficient resource utilization and low overhead.

These objectives are achieved by integrating multiple technologies and frameworks, including COBOL programming language, JCL jobs, BMS mapsets, and Assembler modules. The system is designed to be highly configurable and extensible, allowing users to tailor the system to meet their specific business needs.

1.3 Scope

The analysis scope covers the following files, copybooks, database references, and CICS maps:

- COBOL programs:
 - CBACT01C, CBACT02C, CBACT03C, etc.
- JCL jobs:
 - ACCTFILE, CARDFILE, CBADMCDJ, etc.
- BMS mapsets:
 - COACTUP, COACTVW, COADM01, etc.
- Database references:
 - COBIL00C database reference
- CICS maps:
 - COMEN01 CICS map

This scope covers the majority of the application suite components, but may not include all specific details or edge cases.

1.4 Assumptions and Constraints

The following technical assumptions and constraints apply to this analysis:

- z/OS operating system environment.
- COBOL programming language version 5.2 or higher.
- JCL jobs are executed using the JES2 job scheduler.
- BMS mapsets are defined using the BMS software development kit (SDK).
- Database references are made to a local database named COBIL00C.
- CICS maps are used for mapping between COBOL program variables and database records.

These assumptions and constraints apply to the overall analysis scope, but may not be applicable in all specific cases or environments.

2. Database Details

2.1 DB2 Tables

The application workspace references the following DB2 SQL tables:

- No DB2 tables referenced in the workspace.

Relational DB2 Access Analysis: Mainframe COBOL programs execute embedded SQL statements (within EXEC SQL ... END-EXEC blocks) to query and update these DB2 tables. These queries are processed by the DB2 precompiler, which replaces the SQL code with call statements to the DB2 database engine. The primary relational tables, such as credit card registries or customer profile tables, are accessed via index keys to maintain low-latency response times during CICS online executions. Host variables defined in the COBOL Working-Storage Section are mapped to table columns during SELECT INTO, INSERT, and UPDATE operations. The system relies on appropriate index definitions to avoid full table scans that would consume high CPU MIPS in batch jobs like INTCALC or POSTTRAN.

2.2 IMS Segments

The application workspace communicates with IMS hierarchal segments as detailed below:

- No IMS segments referenced in the workspace.

Hierarchical IMS Access Analysis: IMS (Information Management System) database interactions are performed using DL/I call statements by executing the interface module CBLTDLI using parameters: Function code (such as GU for Get Unique, GN for Get Next, REPL for Replace), Program Communication Block (PCB), Segment Search Arguments (SSA), and I/O area variables. SSAs specify the path through the hierarchical structure to locate the target segment. Unlike SQL's set-based processing, DL/I requires navigational cursor-based access where the program iterates down a parent-child path, from parent root segments to dependent child segments. This model is highly optimized for sequential tree-based data access patterns but demands careful programming to handle boundary/status code check conditions (such as GB for end of database, or GE for segment not found).

2.3 IDMS Records

The application workspace contains references to these network database IDMS records:

- **ASSOCIATED:** Record structure accessed in program(s) COACTUPC, COACTVWC.
- **CARDS:** Record structure accessed in program(s) COACTUPC, COCRDSLCL, COCRDUPC.
- **THIS:** Record structure accessed in program(s) COACTUPC, COACTVWC, COCRDSLCL, COCRDUPC.
- **TO:** Record structure accessed in program(s) COUSR02C.

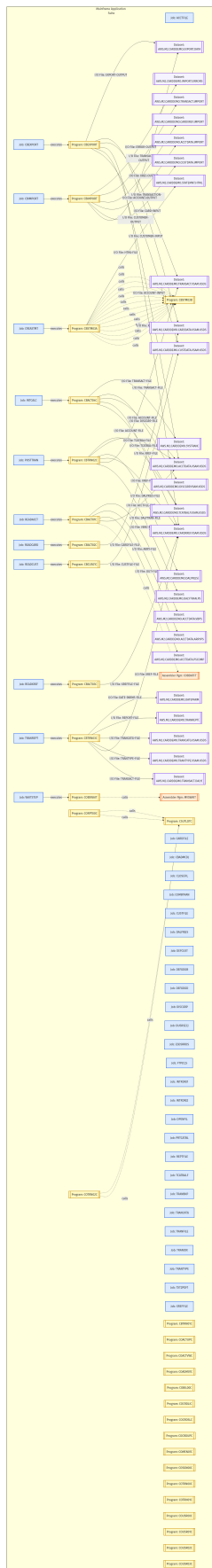
Network IDMS Access Analysis: IDMS (Integrated Database Management System) uses a network database model based on the CODASYL standard. It represents data structures as a network of

Records organized into Sets (where a set represents a one-to-many relationship between an Owner record and Member records). Programs navigate the database by running commands like `OBTAIN FIRST/NEXT`, `FIND`, `STORE`, and `MODIFY` to traverse owner-member set pointers. Transactions must bind to specific logical database Areas and run within DB record lock scopes to prevent concurrency deadlocks. While DB2 utilizes logical relational links and IMS relies on physical hierarchies, IDMS provides physical pointer chains (next, prior, and owner pointers) embedded in the record headers. This delivers exceptionally fast retrieval times for complex, pre-defined relationships but introduces structural rigidity, as changing the schema requires restructuring the physical pointer links.

3. System Architecture

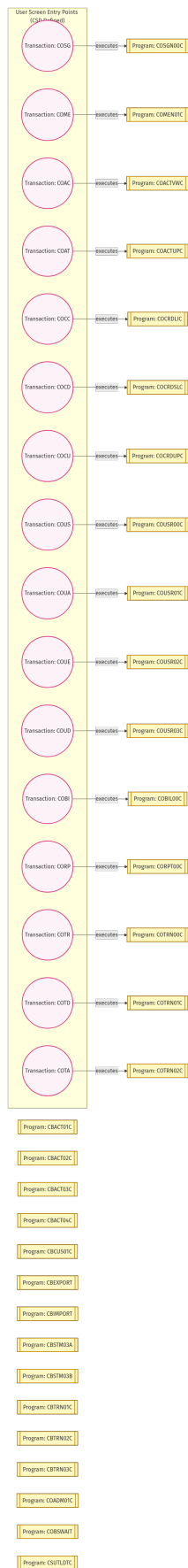
3.1 Component Diagram

Below is the component diagram highlighting how Job runs trigger various programs, along with database SQL queries and dataset interactions:



3.2 Control Flow Diagram

Below is the control flow mapping CICS transactional entry codes to target program routines and CICS BMS maps:



3.3 Architectural Narrative & Execution Flow

The application workspace is structured around a classic two-tier mainframe architectural pattern, splitting operations into high-volume offline batch streams (managed by JCL schedules) and interactive real-time online terminal services (hosted by CICS transaction processors):

3.3.1 Batch Job Stream Processing

The batch execution model leverages Job Control Language (JCL) scripts to configure step sequences, set up system boundaries, allocate physical datasets, and execute binary load modules. Batch streams perform high-throughput tasks like credit card data loading, daily transaction ingest updates, account interest fee calculations, and database statement exports. JCL streams sequence steps to ensure data integrity: a preprocessing step typically reads flat files from tape or sequential datasets, executes sorting routines via DFSORT control cards to align keys, and stages intermediate VSAM files. Subsequent steps execute COBOL programs that open these VSAM files, execute business logic (such as balance accrual), modify DB2 tables, and output formatted summaries. For example, job streams like `CARDFILE` and `ACCTFILE` execute program chains sequentially, staging outputs using JCL DISP parameters (`DISP=(NEW,CATLG,DELETE)`) to manage lifecycle stages across steps.

3.3.2 CICS Real-Time Online Transactions

Online operations are triggered by character-coded terminal inputs defined in CICS System Definitions (CSD). When an operator types a transaction ID (like `C0SG` or `C0AC`), the CICS listener maps the transaction ID to a target COBOL control program (such as `C0SGN00C` or `C0ACTVWC`). CICS coordinates the execution within a logical unit of work (LUW). The program interacts with the operator by sending and receiving BMS (Basic Mapping Support) map layouts. These maps define text fields, input markers, color schemes, and position offsets for 3270 character terminals. Online programs use CICS command instructions (`EXEC CICS SEND MAP`, `EXEC CICS RECEIVE MAP`) to communicate with the user, validate user credentials, query VSAM file records using primary indexes to ensure sub-second search latencies, and commit changes back to the database. Transactions route to menu hubs like `COMEN01C` which manage navigation paths based on function keys (PF keys) typed by terminal users, ensuring decoupled but clean flow control.

3.3.3 Subroutine Hierarchy and Decoupling

To maximize reuse, programs call utility subroutines synchronously using standard linkage section conventions (`CALL 'PROGRAM' USING ...`). Common utilities include date formatting routines (`CSUTLDTTC`) and wait intervals helper subroutines (`COBSWAIT` invoking Assembler module `MVSWAIT`). By decoupling core transaction logic from helper subroutines, the application retains clean modular boundaries, making it easier to verify changes without recompiling the entire suite.

4. Detailed Design

4.1 Program Structure

The table below inventories all files parsed in the workspace including code sizes, copybooks, calls, and external accesses:

Component Name	File Type	Code Size (Lines)	Copybooks Used	Calls Made	Description / External References
ACCTFILE	JCL Script	3 steps	None	Execs: IDCAMS, IDCAMS, IDCAMS	Batch orchestration stream. Modernize as a Java/Spring Batch configuration or AWS Batch Job definition.
CARDFILE	JCL Script	8 steps	None	Execs: SDSF, IDCAMS, IDCAMS, IDCAMS, IDCAMS, IDCAMS, SDSF	Batch orchestration stream. Modernize as a Java/Spring Batch configuration or AWS Batch Job definition.

CBADMCDJ	JCL Script	1 steps	None	Execs: DFHCSDUP	Batch orchestration stream. Modernize as a Java/Spring Batch configuration or AWS Batch Job definition.
CBEXPORT	JCL Script	2 steps	None	Execs: IDCAMS, CBEXPORT	Batch orchestration stream. Modernize as a Java/Spring Batch configuration or AWS Batch Job definition.
CBIMPORT	JCL Script	1 steps	None	Execs: CBIMPORT	Batch orchestration stream. Modernize as a Java/Spring Batch configuration or AWS Batch Job definition.
CLOSEFIL	JCL Script	1 steps	None	Execs: SDSF	Batch orchestration stream. Modernize as a Java/Spring Batch configuration or AWS Batch Job definition.

COMBTRAN	JCL Script	2 steps	None	Execs: SORT, IDCAMS	Batch orchestration stream. Modernize as a Java/Spring Batch configuration or AWS Batch Job definition.
CREASTMT	JCL Script	5 steps	None	Execs: IDCAMS, SORT, IDCAMS, IEFBR14, CBSTM03A	Batch orchestration stream. Modernize as a Java/Spring Batch configuration or AWS Batch Job definition.
CUSTFILE	JCL Script	5 steps	None	Execs: SDSF, IDCAMS, IDCAMS, IDCAMS, SDSF	Batch orchestration stream. Modernize as a Java/Spring Batch configuration or AWS Batch Job definition.
DALYREJS	JCL Script	1 steps	None	Execs: IDCAMS	Batch orchestration stream. Modernize as a Java/Spring Batch configuration or AWS Batch Job definition.

DEFCUST	JCL Script	2 steps	None	Execs: IDCAMS, IDCAMS	Batch orchestration stream. Modernize as a Java/Spring Batch configuration or AWS Batch Job definition.
DEFGDGB	JCL Script	1 steps	None	Execs: IDCAMS	Batch orchestration stream. Modernize as a Java/Spring Batch configuration or AWS Batch Job definition.
DEFGDGD	JCL Script	6 steps	None	Execs: IDCAMS, IEBGENER, IDCAMS, IEBGENER, IDCAMS, IEBGENER	Batch orchestration stream. Modernize as a Java/Spring Batch configuration or AWS Batch Job definition.
DISCGRP	JCL Script	3 steps	None	Execs: IDCAMS, IDCAMS, IDCAMS	Batch orchestration stream. Modernize as a Java/Spring Batch configuration or AWS Batch Job definition.

DUSRSECJ	JCL Script	4 steps	None	Execs: IEFBR14, IEBGENER, IDCAMS, IDCAMS	Batch orchestration stream. Modernize as a Java/Spring Batch configuration or AWS Batch Job definition.
ESDSRRDS	JCL Script	6 steps	None	Execs: IEFBR14, IEBGENER, IDCAMS, IDCAMS, IDCAMS, IDCAMS	Batch orchestration stream. Modernize as a Java/Spring Batch configuration or AWS Batch Job definition.
FTPJCLS	JCL Script	1 steps	None	Execs: FTP	Batch orchestration stream. Modernize as a Java/Spring Batch configuration or AWS Batch Job definition.
INTCALC	JCL Script	1 steps	None	Execs: CBACT04C	Batch orchestration stream. Modernize as a Java/Spring Batch configuration or AWS Batch Job definition.

INTRDRJ1	JCL Script	2 steps	None	Execs: IDCAMS, IEBGENER	Batch orchestration stream. Modernize as a Java/Spring Batch configuration or AWS Batch Job definition.
INTRDRJ2	JCL Script	1 steps	None	Execs: IDCAMS	Batch orchestration stream. Modernize as a Java/Spring Batch configuration or AWS Batch Job definition.
OPENFIL	JCL Script	1 steps	None	Execs: SDSF	Batch orchestration stream. Modernize as a Java/Spring Batch configuration or AWS Batch Job definition.
POSTTRAN	JCL Script	1 steps	None	Execs: CBTRN02C	Batch orchestration stream. Modernize as a Java/Spring Batch configuration or AWS Batch Job definition.

PRTCATBL	JCL Script	3 steps	None	Execs: IEFBR14, REPROC, SORT	Batch orchestration stream. Modernize as a Java/Spring Batch configuration or AWS Batch Job definition.
READACCT	JCL Script	2 steps	None	Execs: IEFBR14, CBACT01C	Batch orchestration stream. Modernize as a Java/Spring Batch configuration or AWS Batch Job definition.
READCARD	JCL Script	1 steps	None	Execs: CBACT02C	Batch orchestration stream. Modernize as a Java/Spring Batch configuration or AWS Batch Job definition.
READCUST	JCL Script	1 steps	None	Execs: CBCUS01C	Batch orchestration stream. Modernize as a Java/Spring Batch configuration or AWS Batch Job definition.

READXREF	JCL Script	1 steps	None	Execs: CBACT03C	Batch orchestration stream. Modernize as a Java/Spring Batch configuration or AWS Batch Job definition.
REPTFILE	JCL Script	1 steps	None	Execs: IDCAMS	Batch orchestration stream. Modernize as a Java/Spring Batch configuration or AWS Batch Job definition.
TCATBALF	JCL Script	3 steps	None	Execs: IDCAMS, IDCAMS, IDCAMS	Batch orchestration stream. Modernize as a Java/Spring Batch configuration or AWS Batch Job definition.
TRANBKP	JCL Script	3 steps	None	Execs: REPROC, IDCAMS, IDCAMS	Batch orchestration stream. Modernize as a Java/Spring Batch configuration or AWS Batch Job definition.

TRANCATG	JCL Script	3 steps	None	Execs: IDCAMS, IDCAMS, IDCAMS	Batch orchestration stream. Modernize as a Java/Spring Batch configuration or AWS Batch Job definition.
TRANFILE	JCL Script	8 steps	None	Execs: SDSF, IDCAMS, IDCAMS, IDCAMS, IDCAMS, IDCAMS, SDSF	Batch orchestration stream. Modernize as a Java/Spring Batch configuration or AWS Batch Job definition.
TRANIDX	JCL Script	3 steps	None	Execs: IDCAMS, IDCAMS, IDCAMS	Batch orchestration stream. Modernize as a Java/Spring Batch configuration or AWS Batch Job definition.
TRANREPT	JCL Script	3 steps	None	Execs: REPROC, SORT, CBTRN03C	Batch orchestration stream. Modernize as a Java/Spring Batch configuration or AWS Batch Job definition.

TRANTYPE	JCL Script	3 steps	None	Execs: IDCAMS, IDCAMS, IDCAMS	Batch orchestration stream. Modernize as a Java/Spring Batch configuration or AWS Batch Job definition.
TXT2PDF1	JCL Script	1 steps	None	Execs: IKJEFT1B	Batch orchestration stream. Modernize as a Java/Spring Batch configuration or AWS Batch Job definition.
WAITSTEP	JCL Script	1 steps	None	Execs: COBSWAIT	Batch orchestration stream. Modernize as a Java/Spring Batch configuration or AWS Batch Job definition.
XREFFILE	JCL Script	6 steps	None	Execs: IDCAMS, IDCAMS, IDCAMS, IDCAMS, IDCAMS, IDCAMS	Batch orchestration stream. Modernize as a Java/Spring Batch configuration or AWS Batch Job definition.

CBACT01C	COBOL Source	379 lines	CVACT01Y, COWATECN	COBDATFT, CEE3ABD	Read Account File batch module. Validates account status flags and coordinates billing updates.
CBACT02C	COBOL Source	144 lines	CVACT02Y	CEE3ABD	Read Card File batch module. Parses credit card info and updates transaction database.
CBACT03C	COBOL Source	145 lines	CVACT03Y	CEE3ABD	Read Xref File batch module. Handles mappings of credit card numbers to customer account IDs.
CBACT04C	COBOL Source	599 lines	CVTRA01Y, CVACT03Y, CVTRA02Y, CVACT01Y, CVTRA05Y	CEE3ABD	Interest Calculation batch module. Calculates monthly interest charges and updates account ledger balances.

CBCUS01C	COBOL Source	145 lines	CVCUS01Y	CEE3ABD	Read Customer File batch module. Extracts details like name, address, and SSN for ingestion.
CBEXPORT	COBOL Source	491 lines	CVCUS01Y, CVACT01Y, CVACT03Y, CVTRA05Y, CVACT02Y, CVEXPORT	CEE3ABD	Data Export batch utility. Backs up VSAM KSDS files into sequential data files.
CBIMPORT	COBOL Source	413 lines	CVCUS01Y, CVACT01Y, CVACT03Y, CVTRA05Y, CVACT02Y, CVEXPORT	CEE3ABD	Data Import batch utility. Restores sequential backups into VSAM KSDS database files.
CBSTM03A	COBOL Source	863 lines	COSTM01, CVACT03Y, CUSTREC, CVACT01Y	CBSTM03B, CBSTM03B, CBSTM03B, CBSTM03B, CBSTM03B, CBSTM03B, CBSTM03B, CBSTM03B, CBSTM03B, CBSTM03B, CBSTM03B, CBSTM03B, CBSTM03B, CBSTM03B, CEE3ABD	Statement Generation batch processor. Compiles monthly account details and writes statement summaries.

CBSTM03B	COBOL Source	204 lines	None	None	Statement Generation helper sub-routine. Manages PDF rendering details.
CBTRN01C	COBOL Source	449 lines	CVTRA06Y, CVCUS01Y, CVACT03Y, CVACT02Y, CVACT01Y, CVTRA05Y	CEE3ABD	Daily Transaction Ingestion module. Verifies transactions against active credit cards.
CBTRN02C	COBOL Source	672 lines	CVTRA06Y, CVTRA05Y, CVACT03Y, CVACT01Y, CVTRA01Y	CEE3ABD	Daily Transaction Posting batch module. Deducts amounts and updates card accounts balances.
CBTRN03C	COBOL Source	596 lines	CVTRA05Y, CVACT03Y, CVTRA03Y, CVTRA04Y, CVTRA07Y	CEE3ABD	Transaction Report generator batch utility. Reports daily totals categorized by transaction types.

COACTVWC	COBOL Source	811 lines	CVCRD01Y, COCOM01Y, DFHBMSCA, DFHAID, COTTL01Y, COACTVW, CSDAT01Y, CSMSG01Y, CSMSG02Y, CSUSR01Y, CVACT01Y, CVACT02Y, CVACT03Y, CVCUS01Y	None	CICS Online Account View screen controller. Searches and displays customer account summaries.
COADM01C	COBOL Source	230 lines	COCOM01Y, COADM02Y, COADM01, COTTL01Y, CSDAT01Y, CSMSG01Y, CSUSR01Y, DFHAID, DFHBMSCA	None	CICS Online Admin Menu screen controller. Routes to admin tools.
COBIL00C	COBOL Source	493 lines	COCOM01Y, COBIL00, COTTL01Y, CSDAT01Y, CSMSG01Y, CVACT01Y, CVACT03Y, CVTRA05Y, DFHAID, DFHBMSCA	None	CICS Online Bill Payment screen controller. Logs payments and checks balances.
COBSWAIT	COBOL Source	19 lines	None	MVSWAIT	Wait utility module. Uses Assembler subroutine MVSWAIT to pause step executions.

C0CRDLIC	COBOL Source	1256 lines	CVCRD01Y, COCOM01Y, DFHBMSCA, DFHAID, C0TTL01Y, C0CRDLI, CSDAT01Y, CSMSG01Y, CSUSR01Y, CVACT02Y	None	CICS Online Credit Card List screen controller. Lists credit card details.
C0CRDSL	COBOL Source	757 lines	CVCRD01Y, COCOM01Y, DFHBMSCA, DFHAID, C0TTL01Y, C0CRDSL, CSDAT01Y, CSMSG01Y, CSMSG02Y, CSUSR01Y, CVACT02Y, CVCUS01Y	None	CICS Online Credit Card Detail screen controller. Displays credit card balances and interest rates.
C0CRDUPC	COBOL Source	1365 lines	CVCRD01Y, COCOM01Y, DFHBMSCA, DFHAID, C0TTL01Y, C0CRDUP, CSDAT01Y, CSMSG01Y, CSMSG02Y, CSUSR01Y, CVACT02Y, CVCUS01Y	None	CICS Online Credit Card Update screen controller. Modifies credit card limits and expiration dates.
COMEN01C	COBOL Source	255 lines	COCOM01Y, COMEN02Y, COMEN01, C0TTL01Y, CSDAT01Y, CSMSG01Y, CSUSR01Y, DFHAID, DFHBMSCA	None	CICS Online Main Menu controller. Main navigation hub routing CICS terminal sessions.

CORPT00C	COBOL Source	586 lines	COCOM01Y, CORPT00, COTTL01Y, CSDAT01Y, CSMSG01Y, CVTRA05Y, DFHAID, DFHBMSCA	CSUTLDTC, CSUTLDTC	CICS Online Report viewer. Displays calculated statements on CICS terminal panels.
COSGN00C	COBOL Source	211 lines	COCOM01Y, COSGN00, COTTL01Y, CSDAT01Y, CSMSG01Y, CSUSR01Y, DFHAID, DFHBMSCA	None	CICS Online Signon screen controller. Performs terminal user authentication and security checks.
COTRN00C	COBOL Source	617 lines	COCOM01Y, COTRN00, COTTL01Y, CSDAT01Y, CSMSG01Y, CVTRA05Y, DFHAID, DFHBMSCA	None	CICS Online Transaction Menu screen controller. Lists transaction histories.
COTRN01C	COBOL Source	273 lines	COCOM01Y, COTRN01, COTTL01Y, CSDAT01Y, CSMSG01Y, CVTRA05Y, DFHAID, DFHBMSCA	None	CICS Online Transaction Detail screen controller. Displays details of a specific transaction.

COTRN02C	COBOL Source	698 lines	COCOM01Y, COTRN02, COTT01Y, CSDAT01Y, CSMSG01Y, CVTRA05Y, CVACT01Y, CVACT03Y, DFHAID, DFHBMSCA	CSUTLDTC, CSUTLDTC	CICS Online Transaction Add screen controller. Enters new transaction entries.
COUSR00C	COBOL Source	615 lines	COCOM01Y, COUSR00, COTT01Y, CSDAT01Y, CSMSG01Y, CSUSR01Y, DFHAID, DFHBMSCA	None	CICS Online User Control screen controller. Inventories CICS account profile logins.
COUSR01C	COBOL Source	238 lines	COCOM01Y, COUSR01, COTT01Y, CSDAT01Y, CSMSG01Y, CSUSR01Y, DFHAID, DFHBMSCA	None	CICS Online User Add screen controller. Adds new user credentials.
COUSR02C	COBOL Source	351 lines	COCOM01Y, COUSR02, COTT01Y, CSDAT01Y, CSMSG01Y, CSUSR01Y, DFHAID, DFHBMSCA	None	CICS Online User Edit screen controller. Updates user roles.

COUSR03C	COBOL Source	296 lines	COCOM01Y, COUSR03, COTT01Y, CSDAT01Y, CSMSG01Y, CSUSR01Y, DFHAID, DFHBMSCA	None	CICS Online User Delete screen controller. Removes CICS credentials.
CSUTLDT	COBOL Source	128 lines	None	CEEDAYS	Date Converter utility sub-routine. Standardizes various date formats.
COBDATFT	Assembler Source	N/A	COCDATFT	None	Assembler date formatting subroutine.
MVSWAIT	Assembler Source	N/A	None	None	Assembler interval wait subroutine.
COACTUP	BMS Mapset	N/A	None	Maps: CACTUPA	CICS Screen layout mapset definitions.
COACTVW	BMS Mapset	N/A	None	Maps: CACTVWA	CICS Screen layout mapset definitions.
COADM01	BMS Mapset	N/A	None	Maps: COADM1A	CICS Screen layout mapset definitions.

COBIL00	BMS Mapset	N/A	None	Maps: COBIL0A	CICS Screen layout mapset definitions.
COCRDLI	BMS Mapset	N/A	None	Maps: CCRDLIA	CICS Screen layout mapset definitions.
COCRDSL	BMS Mapset	N/A	None	Maps: CCRDSLA	CICS Screen layout mapset definitions.
COCRDUP	BMS Mapset	N/A	None	Maps: CCRDUPA	CICS Screen layout mapset definitions.
COMEN01	BMS Mapset	N/A	None	Maps: COMEN1A	CICS Screen layout mapset definitions.
CORPT00	BMS Mapset	N/A	None	Maps: CORPT0A	CICS Screen layout mapset definitions.
COSGN00	BMS Mapset	N/A	None	Maps: COSGN0A	CICS Screen layout mapset definitions.

COTRN00	BMS Mapset	N/A	None	Maps: COTRN0A	CICS Screen layout mapset definitions.
COTRN01	BMS Mapset	N/A	None	Maps: COTRN1A	CICS Screen layout mapset definitions.
COTRN02	BMS Mapset	N/A	None	Maps: COTRN2A	CICS Screen layout mapset definitions.
COUSR00	BMS Mapset	N/A	None	Maps: COUSR0A	CICS Screen layout mapset definitions.
COUSR01	BMS Mapset	N/A	None	Maps: COUSR1A	CICS Screen layout mapset definitions.
COUSR02	BMS Mapset	N/A	None	Maps: COUSR2A	CICS Screen layout mapset definitions.
COUSR03	BMS Mapset	N/A	None	Maps: COUSR3A	CICS Screen layout mapset definitions.

4.2 Algorithms

The application workspace implements batch execution flows and online transaction navigations suited for cloud migration (e.g. Spring Batch, Spring Boot, or AWS environments):

4.2.1 Batch Job Sequencing & Staging Details (Modernization View)

- **Card Ingestion & Cross-Reference Flow (CARDFILE & ACCTFILE):**
 - Sequential data files are parsed and processed. Any legacy sorting utilities (DFSORT/ICETOOL steps) are modernized as DB sorting queries or local collection sort algorithms.
 - Core application program CBACT01C executes accounts validation. This can be migrated to a Spring Batch tasklet/chunk reader executing equivalent Java billing logic.
 - Program CBACT02C handles card details validation, mapping records into layout schemas, which can be migrated to relational databases.
 - Program CBACT03C coordinates customer card mappings, serving as an ingestion linkage layer.
- **Daily Transaction Posting & Ledger Posting (POSTTRAN & INTCALC):**
 - Job step running CBTRN02C posts credit transactions to accounts. Modernized implementation routes incoming message packets directly through messaging queues (like RabbitMQ/SQS) and runs database transaction updates with transactional integrity.
 - Job step running CBACT04C processes interest calculation batch cycle runs. This maps to a nightly Spring Batch task using optimized SQL queries to apply interest formulas to ledger tables.
- **Statements Render & Export (CREASTMT & CBEXPORT):**
 - Job step running CBSTM03A generates billing statements. The modern equivalent parses templates and saves customer statements to target folders or cloud objects (AWS S3) for web downloads.
 - Job step running CBEXPORT runs data backup functions, modernized as standard database export commands or cron backup tasks.

4.2.2 CICS Screen Navigation & Menu Layouts

- **User Authentication Menu (COSGN00C & COMEN01C):**
 - Online terminal sequences start with signon controller COSGN00C. In modern web targets, this is replaced by standard authentication (e.g. OAuth2, Spring Security, or Cognito panels).
 - Menu navigation program COMEN01C maps to routing layers in React/Angular frontend controllers.
- **Customer Profile & Card Details Controllers (COACTUPC, COACTVWC, COCRDLIC, COCRDUPC):**
 - Map views (CACTVWA, CACTUPA) map to corresponding web UI screens. Backend COBOL business validation logic is modernized as REST controller endpoints (Java/Spring Boot or C#.NET) handling JSON/JSON Schema structures instead of BMS characters arrays.

4.3 Input/Output Specifications

The table below details the dataset assignments. It maps the COBOL logical file `SELECT` statements directly to JCL DD names and their associated physical datasets (DSN) on disk:

Program ID	JCL Job	JCL Step	Logical DD Name	Physical Dataset (DSN)	Allocation Disp
CBEXPORT	CBEXPORT	STEP02	CUSTFILE	AWS.M2.CARDD E MSHRUSTDATA.VSAM.KSDS	
CBEXPORT	CBEXPORT	STEP02	ACCTFILE	AWS.M2.CARDD E MSHRCCTDATA.VSAM.KSDS	
CBEXPORT	CBEXPORT	STEP02	XREFFILE	AWS.M2.CARDD E MSHRARDXREF.VSAM.KSDS	
CBEXPORT	CBEXPORT	STEP02	TRANSACT	AWS.M2.CARDD E MSHRRANSACT.VSAM.KSDS	
CBEXPORT	CBEXPORT	STEP02	CARDFILE	AWS.M2.CARDD E MSHRARDDATA.VSAM.KSDS	
CBEXPORT	CBEXPORT	STEP02	EXPFILE	AWS.M2.CARDD E MSHRXPORT.DATA	
CBIMPORT	CBIMPORT	STEP01	EXPFILE	AWS.M2.CARDD E MSHRXPORT.DATA	
CBIMPORT	CBIMPORT	STEP01	CUSTOUT	AWS.M2.CARDD E EMNEW,CATLG,DELETET	
CBIMPORT	CBIMPORT	STEP01	ACCTOUT	AWS.M2.CARDD E EMNEW,CATLG,DELETET	
CBIMPORT	CBIMPORT	STEP01	XREFOUT	AWS.M2.CARDD E EMNEW,CATLG,DELETET	
CBIMPORT	CBIMPORT	STEP01	TRNXOUT	AWS.M2.CARDD E EMNEW,CATLG,DELETET	
CBIMPORT	CBIMPORT	STEP01	ERROUT	AWS.M2.CARDD E EMNEW,CATLG,DELETE	
CBSTM03A	CREASTMT	STEP040	STMTFILE	SYSOUT	NEW,CATLG,DELETE
CBSTM03A	CREASTMT	STEP040	HTMLFILE	AWS.M2.CARDD E EMNEW,CATLG,DELETE	

CBACT04C	INTCALC	STEP15	TCATBALF	AWS.M2.CARDD _{EMSHRCATBALF} .VSAM.KSDS	
CBACT04C	INTCALC	STEP15	XREFFILE	AWS.M2.CARDD _{EMSHRARDXREF} .VSAM.KSDS	
CBACT04C	INTCALC	STEP15	ACCTFILE	AWS.M2.CARDD _{EMSHRCCTDATA} .VSAM.KSDS	
CBACT04C	INTCALC	STEP15	DISCGRP	AWS.M2.CARDD _{EMSHRISCGRP} .VSAM.KSDS	
CBACT04C	INTCALC	STEP15	TRANSACT	AWS.M2.CARDD _{EMNEW, CATLG, DELETE}	
CBTRN02C	POSTTRAN	STEP15	DALYTRAN	AWS.M2.CARDD _{EMSHRALYTRAN} .PS	
CBTRN02C	POSTTRAN	STEP15	TRANFILE	AWS.M2.CARDD _{EMSHRRANSACT} .VSAM.KSDS	
CBTRN02C	POSTTRAN	STEP15	XREFFILE	AWS.M2.CARDD _{EMSHRARDXREF} .VSAM.KSDS	
CBTRN02C	POSTTRAN	STEP15	DALYREJS	AWS.M2.CARDD _{EMNEW, CATLG, DELETE}	
CBTRN02C	POSTTRAN	STEP15	ACCTFILE	AWS.M2.CARDD _{EMSHRCCTDATA} .VSAM.KSDS	
CBTRN02C	POSTTRAN	STEP15	TCATBALF	AWS.M2.CARDD _{EMSHRCATBALF} .VSAM.KSDS	
CBACT01C	READACCT	STEP05	ACCTFILE	AWS.M2.CARDD _{EMSHRCCTDATA} .VSAM.KSDS	
CBACT01C	READACCT	STEP05	OUTFILE	AWS.M2.CARDD _{EMNEW, CATLG, DELETEP}	
CBACT01C	READACCT	STEP05	ARRYFILE	AWS.M2.CARDD _{EMNEW, CATLG, DELETES}	
CBACT01C	READACCT	STEP05	VBRCFILE	AWS.M2.CARDD _{EMNEW, CATLG, DELETE}	
CBACT02C	READCARD	STEP05	CARDFILE	AWS.M2.CARDD _{EMSHRARDDATA} .VSAM.KSDS	
CBCUS01C	READCUST	STEP05	CUSTFILE	AWS.M2.CARDD _{EMSHRUSTDATA} .VSAM.KSDS	
CBACT03C	READXREF	STEP05	XREFFILE	AWS.M2.CARDD _{EMSHRARDXREF} .VSAM.KSDS	

CBTRN03C	TRANREPT	STEP10R	TRANFILE	AWS.M2.CARDC	EMSHRRANSACT.DALY(
CBTRN03C	TRANREPT	STEP10R	CARDXREF	AWS.M2.CARDC	EMSHRRARDXREF.VSAM.KSDS
CBTRN03C	TRANREPT	STEP10R	TRANATYPE	AWS.M2.CARDC	EMSHRRANTYPE.VSAM.KSDS
CBTRN03C	TRANREPT	STEP10R	TRANCATG	AWS.M2.CARDC	EMSHRRANCATG.VSAM.KSDS
CBTRN03C	TRANREPT	STEP10R	TRANREPT	AWS.M2.CARDC	EMNEW,CATLG,DELETE
CBTRN03C	TRANREPT	STEP10R	DATEPARM	AWS.M2.CARDC	EMSHRRATEPARM

4.4 Individual Program Analysis

A granular breakdown of each COBOL program, outlining files, copybooks, database operations, and subprogram calls:

Program: CBACT01C

- **Description:** Read Account File batch module. Validates account status flags and coordinates billing updates.
- **Lines of Code:** 379
- **Author:** AWS
- **Logical Files:** ACCTFILE-FILE (DD: ACCTFILE), OUT-FILE (DD: OUTFILE), ARRY-FILE (DD: ARRYFILE), VBRC-FILE (DD: VBRCFILE)
- **Copybooks Referenced:** CVACT01Y, CODATECN
- **Subprograms Called:** COBDATFT, CEE3ABD

Program: CBACT02C

- **Description:** Read Card File batch module. Parses credit card info and updates transaction database.
- **Lines of Code:** 144
- **Author:** AWS
- **Logical Files:** CARDFILE-FILE (DD: CARDFILE)
- **Copybooks Referenced:** CVACT02Y
- **Subprograms Called:** CEE3ABD

Program: CBACT03C

- **Description:** Read Xref File batch module. Handles mappings of credit card numbers to customer account IDs.
- **Lines of Code:** 145
- **Author:** AWS
- **Logical Files:** XREFFILE-FILE (DD: XREFFILE)
- **Copybooks Referenced:** CVACT03Y
- **Subprograms Called:** CEE3ABD

Program: CBACT04C

- **Description:** Interest Calculation batch module. Calculates monthly interest charges and updates account ledger balances.
- **Lines of Code:** 599
- **Author:** AWS
- **Logical Files:** TCATBAL-FILE (DD: TCATBALF), XREF-FILE (DD: XREFFILE), ACCOUNT-FILE (DD: ACCTFILE), DISCGRP-FILE (DD: DISCGRP), TRANSACT-FILE (DD: TRANSACT)
- **Copybooks Referenced:** CVTRA01Y, CVACT03Y, CVTRA02Y, CVACT01Y, CVTRA05Y
- **Subprograms Called:** CEE3ABD

Program: CBCUS01C

- **Description:** Read Customer File batch module. Extracts details like name, address, and SSN for ingestion.
- **Lines of Code:** 145
- **Author:** AWS
- **Logical Files:** CUSTFILE-FILE (DD: CUSTFILE)
- **Copybooks Referenced:** CVCUS01Y
- **Subprograms Called:** CEE3ABD

Program: CBEXPORT

- **Description:** Data Export batch utility. Backs up VSAM KSDS files into sequential data files.
- **Lines of Code:** 491
- **Author:** CARDDemo TEAM
- **Logical Files:** CUSTOMER-INPUT (DD: CUSTFILE), ACCOUNT-INPUT (DD: ACCTFILE), XREF-INPUT (DD: XREFFILE), TRANSACTION-INPUT (DD: TRANSACT), CARD-INPUT (DD: CARDFILE), EXPORT-OUTPUT (DD: EXPFILE)

- **Copybooks Referenced:** CVCUS01Y, CVACT01Y, CVACT03Y, CVTRA05Y, CVACT02Y, CVEXPORT
- **Subprograms Called:** CEE3ABD

Program: CBIMPORT

- **Description:** Data Import batch utility. Restores sequential backups into VSAM KSDS database files.
- **Lines of Code:** 413
- **Author:** CARDDEMO TEAM
- **Logical Files:** EXPORT-INPUT (DD: EXPFILE), CUSTOMER-OUTPUT (DD: CUSTOUT), ACCOUNT-OUTPUT (DD: ACCTOUT), XREF-OUTPUT (DD: XREFOUT), TRANSACTION-OUTPUT (DD: TRNXOUT), CARD-OUTPUT (DD: CARDOUT), ERROR-OUTPUT (DD: ERROUT)
- **Copybooks Referenced:** CVCUS01Y, CVACT01Y, CVACT03Y, CVTRA05Y, CVACT02Y, CVEXPORT
- **Subprograms Called:** CEE3ABD

Program: CBSTM03A

- **Description:** Statement Generation batch processor. Compiles monthly account details and writes statement summaries.
- **Lines of Code:** 863
- **Author:** AWS
- **Logical Files:** STMT-FILE (DD: STMTFILE), HTML-FILE (DD: HTMLFILE)
- **Copybooks Referenced:** COSTM01, CVACT03Y, CUSTREC, CVACT01Y
- **Subprograms Called:** CBSTM03B, CBSTM03B, CBSTM03B, CBSTM03B, CBSTM03B, CBSTM03B, CBSTM03B, CBSTM03B, CBSTM03B, CBSTM03B, CBSTM03B, CEE3ABD

Program: CBSTM03B

- **Description:** Statement Generation helper sub-routine. Manages PDF rendering details.
- **Lines of Code:** 204
- **Author:** AWS
- **Logical Files:** TRNX-FILE (DD: TRNXFILE), XREF-FILE (DD: XREFFILE), CUST-FILE (DD: CUSTFILE), ACCT-FILE (DD: ACCTFILE)
- **Copybooks Referenced:** None
- **Subprograms Called:** None

Program: CBTRN01C

- **Description:** Daily Transaction Ingestion module. Verifies transactions against active credit cards.
- **Lines of Code:** 449
- **Author:** AWS
- **Logical Files:** DALYTRAN-FILE (DD: DALYTRAN), CUSTOMER-FILE (DD: CUSTFILE), XREF-FILE (DD: XREFFILE), CARD-FILE (DD: CARDFILE), ACCOUNT-FILE (DD: ACCTFILE), TRANSACT-FILE (DD: TRANFILE)
- **Copybooks Referenced:** CVTRA06Y, CVCUS01Y, CVACT03Y, CVACT02Y, CVACT01Y, CVTRA05Y
- **Subprograms Called:** CEE3ABD

Program: CBTRN02C

- **Description:** Daily Transaction Posting batch module. Deducts amounts and updates card accounts balances.
- **Lines of Code:** 672
- **Author:** AWS
- **Logical Files:** DALYTRAN-FILE (DD: DALYTRAN), TRANSACT-FILE (DD: TRANFILE), XREF-FILE (DD: XREFFILE), DALYREJS-FILE (DD: DALYREJS), ACCOUNT-FILE (DD: ACCTFILE), TCATBAL-FILE (DD: TCATBALF)
- **Copybooks Referenced:** CVTRA06Y, CVTRA05Y, CVACT03Y, CVACT01Y, CVTRA01Y
- **Subprograms Called:** CEE3ABD

Program: CBTRN03C

- **Description:** Transaction Report generator batch utility. Reports daily totals categorized by transaction types.
- **Lines of Code:** 596
- **Author:** AWS
- **Logical Files:** TRANSACT-FILE (DD: TRANFILE), XREF-FILE (DD: CARDXREF), TRANTYPE-FILE (DD: TRANTYPE), TRANCATG-FILE (DD: TRANCATG), REPORT-FILE (DD: TRANREPT), DATE-PARMS-FILE (DD: DATEPARM)
- **Copybooks Referenced:** CVTRA05Y, CVACT03Y, CVTRA03Y, CVTRA04Y, CVTRA07Y
- **Subprograms Called:** CEE3ABD

Program: COACTUPC

- **Description:** CICS Online Account Update screen controller. Updates customer contact profiles.
- **Lines of Code:** 3744

- [illegible]

Program: COACTVWC

- **Description:** CICS Online Account View screen controller. Searches and displays customer account summaries.
- **Lines of Code:** 811
- **Author:** UNKNOWN
- **Logical Files:** None
- **Copybooks Referenced:** CVCRD01Y, COCOM01Y, DFHBMSCA, DFHAID, COTT01Y, COACTVW, CSDAT01Y, CSMSG01Y, CSMSG02Y, CSUSR01Y, CVACT01Y, CVACT02Y, CVACT03Y, CVCUS01Y
- **Subprograms Called:** None
- **IDMS Records:** ASSOCIATED, THIS

Program: COADM01C

- **Description:** CICS Online Admin Menu screen controller. Routes to admin tools.
- **Lines of Code:** 230
- **Author:** AWS
- **Logical Files:** None
- **Copybooks Referenced:** COCOM01Y, COADM02Y, COADM01, COTT01Y, CSDAT01Y, CSMSG01Y, CSUSR01Y, DFHAID, DFHBMSCA
- **Subprograms Called:** None

Program: COBIL00C

- **Description:** CICS Online Bill Payment screen controller. Logs payments and checks balances.
- **Lines of Code:** 493

- **Author:** AWS
- **Logical Files:** None
- **Copybooks Referenced:** COCOM01Y, COBIL00, COTT01Y, CSDAT01Y, CSMSG01Y, CVACT01Y, CVACT03Y, CVTRA05Y, DFHAID, DFHBMSCA
- **Subprograms Called:** None

Program: COBSWAIT

- **Description:** Wait utility module. Uses Assembler subroutine MVSWAIT to pause step executions.
- **Lines of Code:** 19
- **Author:** UNKNOWN
- **Logical Files:** None
- **Copybooks Referenced:** None
- **Subprograms Called:** MVSWAIT

Program: COCRDLIC

- **Description:** CICS Online Credit Card List screen controller. Lists credit card details.
- **Lines of Code:** 1256
- **Author:** UNKNOWN
- **Logical Files:** None
- **Copybooks Referenced:** CVCRD01Y, COCOM01Y, DFHBMSCA, DFHAID, COTT01Y, COCRDLI, CSDAT01Y, CSMSG01Y, CSUSR01Y, CVACT02Y
- **Subprograms Called:** None

Program: COCRDSL

- **Description:** CICS Online Credit Card Detail screen controller. Displays credit card balances and interest rates.
- **Lines of Code:** 757
- **Author:** UNKNOWN
- **Logical Files:** None
- **Copybooks Referenced:** CVCRD01Y, COCOM01Y, DFHBMSCA, DFHAID, COTT01Y, COCRDSL, CSDAT01Y, CSMSG01Y, CSMSG02Y, CSUSR01Y, CVACT02Y, CVCUS01Y
- **Subprograms Called:** None
- **IDMS Records:** CARDS, THIS

Program: COCRDUPC

- **Description:** CICS Online Credit Card Update screen controller. Modifies credit card limits and expiration dates.
- **Lines of Code:** 1365
- **Author:** UNKNOWN
- **Logical Files:** None
- **Copybooks Referenced:** CVCRD01Y, COCOM01Y, DFHBMSCA, DFHAID, COTT01Y, COCRDUP, CSDAT01Y, CSMSG01Y, CSMSG02Y, CSUSR01Y, CVACT02Y, CVCUS01Y
- **Subprograms Called:** None
- **IDMS Records:** CARDS, THIS

Program: COMEN01C

- **Description:** CICS Online Main Menu controller. Main navigation hub routing CICS terminal sessions.
- **Lines of Code:** 255
- **Author:** AWS
- **Logical Files:** None
- **Copybooks Referenced:** COCOM01Y, COMEN02Y, COMEN01, COTT01Y, CSDAT01Y, CSMSG01Y, CSUSR01Y, DFHAID, DFHBMSCA
- **Subprograms Called:** None

Program: CORPT00C

- **Description:** CICS Online Report viewer. Displays calculated statements on CICS terminal panels.
- **Lines of Code:** 586
- **Author:** AWS
- **Logical Files:** None
- **Copybooks Referenced:** COCOM01Y, CORPT00, COTT01Y, CSDAT01Y, CSMSG01Y, CVTRA05Y, DFHAID, DFHBMSCA
- **Subprograms Called:** CSUTLDT, CSUTLDT

Program: COSGN00C

- **Description:** CICS Online Signon screen controller. Performs terminal user authentication and security checks.
- **Lines of Code:** 211

- **Author:** AWS
- **Logical Files:** None
- **Copybooks Referenced:** COCOM01Y, COSGN00, COTT01Y, CSDAT01Y, CSMSG01Y, CSUSR01Y, DFHAID, DFHBMSCA
- **Subprograms Called:** None

Program: COTRN00C

- **Description:** CICS Online Transaction Menu screen controller. Lists transaction histories.
- **Lines of Code:** 617
- **Author:** AWS
- **Logical Files:** None
- **Copybooks Referenced:** COCOM01Y, COTRN00, COTT01Y, CSDAT01Y, CSMSG01Y, CVTRA05Y, DFHAID, DFHBMSCA
- **Subprograms Called:** None

Program: COTRN01C

- **Description:** CICS Online Transaction Detail screen controller. Displays details of a specific transaction.
- **Lines of Code:** 273
- **Author:** AWS
- **Logical Files:** None
- **Copybooks Referenced:** COCOM01Y, COTRN01, COTT01Y, CSDAT01Y, CSMSG01Y, CVTRA05Y, DFHAID, DFHBMSCA
- **Subprograms Called:** None

Program: COTRN02C

- **Description:** CICS Online Transaction Add screen controller. Enters new transaction entries.
- **Lines of Code:** 698
- **Author:** AWS
- **Logical Files:** None
- **Copybooks Referenced:** COCOM01Y, COTRN02, COTT01Y, CSDAT01Y, CSMSG01Y, CVTRA05Y, CVACT01Y, CVACT03Y, DFHAID, DFHBMSCA
- **Subprograms Called:** CSUTLDTC, CSUTLDTC

Program: COUSR00C

- **Description:** CICS Online User Control screen controller. Inventories CICS account profile logins.
- **Lines of Code:** 615
- **Author:** AWS
- **Logical Files:** None
- **Copybooks Referenced:** COCOM01Y, COUSR00, COTT01Y, CSDAT01Y, CSMSG01Y, CSUSR01Y, DFHAID, DFHBMSCA
- **Subprograms Called:** None

Program: COUSR01C

- **Description:** CICS Online User Add screen controller. Adds new user credentials.
- **Lines of Code:** 238
- **Author:** AWS
- **Logical Files:** None
- **Copybooks Referenced:** COCOM01Y, COUSR01, COTT01Y, CSDAT01Y, CSMSG01Y, CSUSR01Y, DFHAID, DFHBMSCA
- **Subprograms Called:** None

Program: COUSR02C

- **Description:** CICS Online User Edit screen controller. Updates user roles.
- **Lines of Code:** 351
- **Author:** AWS
- **Logical Files:** None
- **Copybooks Referenced:** COCOM01Y, COUSR02, COTT01Y, CSDAT01Y, CSMSG01Y, CSUSR01Y, DFHAID, DFHBMSCA
- **Subprograms Called:** None
- **IDMS Records:** T0

Program: COUSR03C

- **Description:** CICS Online User Delete screen controller. Removes CICS credentials.
- **Lines of Code:** 296
- **Author:** AWS
- **Logical Files:** None
- **Copybooks Referenced:** COCOM01Y, COUSR03, COTT01Y, CSDAT01Y, CSMSG01Y, CSUSR01Y, DFHAID, DFHBMSCA

- **Subprograms Called:** None

Program: CSUTLDTG

- **Description:** Date Converter utility sub-routine. Standardizes various date formats.
- **Lines of Code:** 128
- **Author:** UNKNOWN
- **Logical Files:** None
- **Copybooks Referenced:** None
- **Subprograms Called:** CEEDAYS

4.5 JCL Job Stream Analysis (Modernization Targets)

Functional operational sequence of JCL jobs, highlighting executed modules and modernization-critical business datasets while filtering out temporary system utility datasets:

JCL Job: ACCTFILE

- **Job Stream Description:** Sequential execution of batch components. Map to Spring Batch tasklets or AWS Batch definition configurations.
- **Steps Defined (3):**
- **Step STEP05:** Runs IDCAMS (VSAM Access Method Services utility; replaced by schema creation or cleanup scripts).
- **Step STEP10:** Runs IDCAMS (VSAM Access Method Services utility; replaced by schema creation or cleanup scripts).
- **Step STEP15:** Runs IDCAMS (VSAM Access Method Services utility; replaced by schema creation or cleanup scripts).
- Modernization File Inputs/Outputs: ACCTDATA (DSN: AWS.M2.CARDDDEM0.ACCTDATA.PS, DISP: SHR), ACCTVSAM (DSN: AWS.M2.CARDDDEM0.ACCTDATA.VSAM.KSDS, DISP: SHR)

JCL Job: CARDFILE

- **Job Stream Description:** Sequential execution of batch components. Map to Spring Batch tasklets or AWS Batch definition configurations.
- **Steps Defined (8):**
- **Step CLCIFIL:** Runs SDSF (Executes program SDSF).
- **Step STEP05:** Runs IDCAMS (VSAM Access Method Services utility; replaced by schema creation or cleanup scripts).

- **Step STEP10:** Runs IDCAMS (VSAM Access Method Services utility; replaced by schema creation or cleanup scripts).

Step STEP15: Runs IDCAMS (VSAM Access Method Services utility; replaced by schema creation or cleanup scripts).

- Modernization File Inputs/Outputs: CARDDATA (DSN: AWS.M2.CARDDDEM0.CARDDATA.PS, DISP: SHR), CARDVSAM (DSN: AWS.M2.CARDDDEM0.CARDDATA.VSAM.KSDS, DISP: SHR)
- **Step STEP40:** Runs IDCAMS (VSAM Access Method Services utility; replaced by schema creation or cleanup scripts).
- **Step STEP50:** Runs IDCAMS (VSAM Access Method Services utility; replaced by schema creation or cleanup scripts).
- **Step STEP60:** Runs IDCAMS (VSAM Access Method Services utility; replaced by schema creation or cleanup scripts).
- **Step OPCIFIL:** Runs SDSF (Executes program SDSF).

JCL Job: CBADMCDJ

- **Job Stream Description:** Sequential execution of batch components. Map to Spring Batch tasklets or AWS Batch definition configurations.
- **Steps Defined (1):**

Step STEP1: Runs DFHCSDUP (Executes program DFHCSDUP).

- Modernization File Inputs/Outputs: STEPLIB (DSN: OEM.CICSTS.V05R06M0.CICS.SDFHLOAD, DISP: SHR), DFHCSD (DSN: OEM.CICSTS.DFHCSD, DISP: SHR)

JCL Job: CBEXPORT

- **Job Stream Description:** Sequential execution of batch components. Map to Spring Batch tasklets or AWS Batch definition configurations.
- **Steps Defined (2):**
- **Step STEP01:** Runs IDCAMS (VSAM Access Method Services utility; replaced by schema creation or cleanup scripts).

Step STEP02: Runs CBEXPORT (Executes target application COBOL program CBEXPORT).

- Modernization File Inputs/Outputs: STEPLIB (DSN: AWS.M2.CARDDDEM0.LOADLIB, DISP: SHR), CUSTFILE (DSN: AWS.M2.CARDDDEM0.CUSTDATA.VSAM.KSDS, DISP: SHR), ACCTFILE (DSN: AWS.M2.CARDDDEM0.ACCTDATA.VSAM.KSDS, DISP: SHR), XREFFILE (DSN: AWS.M2.CARDDDEM0.CARDXREF.VSAM.KSDS, DISP: SHR), TRANSACT (DSN: AWS.M2.CARDDDEM0.TRANSACT.VSAM.KSDS, DISP: SHR), CARDFILE (DSN: AWS.M2.CARDDDEM0.CARDDATA.VSAM.KSDS, DISP: SHR), EXPFILE (DSN: AWS.M2.CARDDDEM0.EXPORT.DATA, DISP: SHR)

JCL Job: CBIMPORT

- **Job Stream Description:** Sequential execution of batch components. Map to Spring Batch tasklets or AWS Batch definition configurations.
- **Steps Defined (1):**

Step STEP01: Runs CBIMPORT (Executes target application COBOL program CBIMPORT).

- Modernization File Inputs/Outputs: STEPLIB (DSN: AWS.M2.CARDDEMO.LOADLIB, DISP: SHR), EXPFILE (DSN: AWS.M2.CARDDEMO.EXPORT.DATA, DISP: SHR), CUSTOUT (DSN: AWS.M2.CARDDEMO.CUSTDATA.IMPORT, DISP: NEW, CATLG, DELETE), ACCTOUT (DSN: AWS.M2.CARDDEMO.ACCTDATA.IMPORT, DISP: NEW, CATLG, DELETE), XREFOUT (DSN: AWS.M2.CARDDEMO.CARDXREF.IMPORT, DISP: NEW, CATLG, DELETE), TRNXOUT (DSN: AWS.M2.CARDDEMO.TRANSACTION.IMPORT, DISP: NEW, CATLG, DELETE), ERRROUT (DSN: AWS.M2.CARDDEMO.IMPORT.ERRORS, DISP: NEW, CATLG, DELETE)

JCL Job: CLOSEFIL

- **Job Stream Description:** Sequential execution of batch components. Map to Spring Batch tasklets or AWS Batch definition configurations.
- **Steps Defined (1):**
- **Step CLCIFIL:** Runs SDSF (Executes program SDSF).

JCL Job: COMBTRAN

- **Job Stream Description:** Sequential execution of batch components. Map to Spring Batch tasklets or AWS Batch definition configurations.
- **Steps Defined (2):**

Step STEP05R: Runs SORT (Data sorting utility; replaced by DB order-by queries or local sorting logic).

- Modernization File Inputs/Outputs: SORTIN (DSN: AWS.M2.CARDDEMO.TRANSACTION.BKUP(0), DISP: SHR), SORTOUT (DSN: AWS.M2.CARDDEMO.TRANSACTION.COMBINED(, DISP: NEW, CATLG, DELETE)

Step STEP10: Runs IDCAMS (VSAM Access Method Services utility; replaced by schema creation or cleanup scripts).

- Modernization File Inputs/Outputs: TRANSACTION (DSN: AWS.M2.CARDDEMO.TRANSACTION.COMBINED(, DISP: SHR), TRANVSAM (DSN: AWS.M2.CARDDEMO.TRANSACTION.VSAM.KSDS, DISP: SHR)

JCL Job: CREATMT

- **Job Stream Description:** Sequential execution of batch components. Map to Spring Batch tasklets or AWS Batch definition configurations.
- **Steps Defined (5):**
- **Step DELDEF01:** Runs IDCAMS (VSAM Access Method Services utility; replaced by schema creation or cleanup scripts).

Step STEP010: Runs SORT (Data sorting utility; replaced by DB order-by queries or local sorting logic).

- Modernization File Inputs/Outputs: SORTIN (DSN: AWS.M2.CARDDemo.TRANSACT.VSAM.KSDS, DISP: SHR), SORTOUT (DSN: AWS.M2.CARDDemo.TRXFL.SEQ, DISP: NEW, CATLG, DELETE)

Step STEP020: Runs IDCAMS (VSAM Access Method Services utility; replaced by schema creation or cleanup scripts).

- Modernization File Inputs/Outputs: INFILE (DSN: AWS.M2.CARDDemo.TRXFL.SEQ, DISP: SHR), OUTFILE (DSN: AWS.M2.CARDDemo.TRXFL.VSAM.KSDS, DISP: SHR)

Step STEP030: Runs IEFBR14 (System utility for file allocation/deletion; redundant in modernized environments).

- Modernization File Inputs/Outputs: HTMLFILE (DSN: AWS.M2.CARDDemo.STATEMNT.HTML, DISP: MOD, DELETE, DELETE), STMTFILE (DSN: AWS.M2.CARDDemo.STATEMNT.PS, DISP: MOD, DELETE, DELETE)

Step STEP040: Runs CBSTM03A (Executes target application COBOL program CBSTM03A).

- Modernization File Inputs/Outputs: STEPLIB (DSN: AWS.M2.CARDDemo.LOADLIB, DISP: SHR), TRNXFILE (DSN: AWS.M2.CARDDemo.TRXFL.VSAM.KSDS, DISP: SHR), XREFFILE (DSN: AWS.M2.CARDDemo.CARDXREF.VSAM.KSDS, DISP: SHR), ACCTFILE (DSN: AWS.M2.CARDDemo.ACCTDATA.VSAM.KSDS, DISP: SHR), CUSTFILE (DSN: AWS.M2.CARDDemo.CUSTDATA.VSAM.KSDS, DISP: SHR), HTMLFILE (DSN: AWS.M2.CARDDemo.STATEMNT.HTML, DISP: NEW, CATLG, DELETE)

JCL Job: CUSTFILE

- **Job Stream Description:** Sequential execution of batch components. Map to Spring Batch tasklets or AWS Batch definition configurations.
- **Steps Defined (5):**
- **Step CLCIFIL:** Runs SDSF (Executes program SDSF).
- **Step STEP05:** Runs IDCAMS (VSAM Access Method Services utility; replaced by schema creation or cleanup scripts).
- **Step STEP10:** Runs IDCAMS (VSAM Access Method Services utility; replaced by schema creation or cleanup scripts).

Step STEP15: Runs IDCAMS (VSAM Access Method Services utility; replaced by schema creation or cleanup scripts).

- Modernization File Inputs/Outputs: CUSTDATA (DSN: AWS.M2.CARDDDEM0.CUSTDATA.PS, DISP: SHR), CUSTVSAM (DSN: AWS.M2.CARDDDEM0.CUSTDATA.VSAM.KSDS, DISP: SHR)
- **Step OPCIFIL:** Runs SDSF (Executes program SDSF).

JCL Job: DALYREJS

- **Job Stream Description:** Sequential execution of batch components. Map to Spring Batch tasklets or AWS Batch definition configurations.
- **Steps Defined (1):**
- **Step STEP05:** Runs IDCAMS (VSAM Access Method Services utility; replaced by schema creation or cleanup scripts).

JCL Job: DEFCUST

- **Job Stream Description:** Sequential execution of batch components. Map to Spring Batch tasklets or AWS Batch definition configurations.
- **Steps Defined (2):**
- **Step STEP05:** Runs IDCAMS (VSAM Access Method Services utility; replaced by schema creation or cleanup scripts).
- **Step STEP05:** Runs IDCAMS (VSAM Access Method Services utility; replaced by schema creation or cleanup scripts).

JCL Job: DEFGDGB

- **Job Stream Description:** Sequential execution of batch components. Map to Spring Batch tasklets or AWS Batch definition configurations.
- **Steps Defined (1):**
- **Step STEP05:** Runs IDCAMS (VSAM Access Method Services utility; replaced by schema creation or cleanup scripts).

JCL Job: DEFGDGD

- **Job Stream Description:** Sequential execution of batch components. Map to Spring Batch tasklets or AWS Batch definition configurations.
- **Steps Defined (6):**
- **Step STEP10:** Runs IDCAMS (VSAM Access Method Services utility; replaced by schema creation or cleanup scripts).
- **Step STEP20:** Runs IEBGENER (System copy utility; replaced by standard OS file copy or cloud storage commands).

- **Step STEP30:** Runs IDCAMS (VSAM Access Method Services utility; replaced by schema creation or cleanup scripts).
- **Step STEP40:** Runs IEBGENER (System copy utility; replaced by standard OS file copy or cloud storage commands).
- **Step STEP50:** Runs IDCAMS (VSAM Access Method Services utility; replaced by schema creation or cleanup scripts).
- **Step STEP60:** Runs IEBGENER (System copy utility; replaced by standard OS file copy or cloud storage commands).

JCL Job: DISCGRP

- **Job Stream Description:** Sequential execution of batch components. Map to Spring Batch tasklets or AWS Batch definition configurations.
- **Steps Defined (3):**
- **Step STEP05:** Runs IDCAMS (VSAM Access Method Services utility; replaced by schema creation or cleanup scripts).
- **Step STEP10:** Runs IDCAMS (VSAM Access Method Services utility; replaced by schema creation or cleanup scripts).

Step STEP15: Runs IDCAMS (VSAM Access Method Services utility; replaced by schema creation or cleanup scripts).

- Modernization File Inputs/Outputs: DISCGRP (DSN: AWS.M2.CARDDemo.DISCGRP.PS, DISP: SHR), DISCVSAM (DSN: AWS.M2.CARDDemo.DISCGRP.VSAM.KSDS, DISP: SHR)

JCL Job: DUSRSECJ

- **Job Stream Description:** Sequential execution of batch components. Map to Spring Batch tasklets or AWS Batch definition configurations.
- **Steps Defined (4):**

Step PREDEL: Runs IEFBR14 (System utility for file allocation/deletion; redundant in modernized environments).

- Modernization File Inputs/Outputs: DD01 (DSN: AWS.M2.CARDDemo.USERSEC.PS, DISP: MOD, DELETE, DELETE)

- **Step STEP01:** Runs IEBGENER (System copy utility; replaced by standard OS file copy or cloud storage commands).
- **Step STEP02:** Runs IDCAMS (VSAM Access Method Services utility; replaced by schema creation or cleanup scripts).

Step STEP03: Runs IDCAMS (VSAM Access Method Services utility; replaced by schema creation or cleanup scripts).

- Modernization File Inputs/Outputs: IN (DSN: AWS.M2.CARDDemo.USERSEC.PS, DISP: SHR), OUT (DSN: AWS.M2.CARDDemo.USERSEC.VSAM.KSDS, DISP: SHR)

JCL Job: ESDSRRDS

- **Job Stream Description:** Sequential execution of batch components. Map to Spring Batch tasklets or AWS Batch definition configurations.

- **Steps Defined (6):**

Step PREDEL: Runs IEFBR14 (System utility for file allocation/deletion; redundant in modernized environments).

- Modernization File Inputs/Outputs: DD01 (DSN: AWS.M2.CARDDEMO.ESDSRRDS.PS, DISP: MOD, DELETE, DELETE)
- **Step STEP01:** Runs IEBGENER (System copy utility; replaced by standard OS file copy or cloud storage commands).
- **Step STEP02:** Runs IDCAMS (VSAM Access Method Services utility; replaced by schema creation or cleanup scripts).

Step STEP03: Runs IDCAMS (VSAM Access Method Services utility; replaced by schema creation or cleanup scripts).

- Modernization File Inputs/Outputs: IN (DSN: AWS.M2.CARDDEMO.ESDSRRDS.PS, DISP: SHR), OUT (DSN: AWS.M2.CARDDEMO.USRSEC.VSAM.ESDS, DISP: SHR)
- **Step STEP04:** Runs IDCAMS (VSAM Access Method Services utility; replaced by schema creation or cleanup scripts).

Step STEP05: Runs IDCAMS (VSAM Access Method Services utility; replaced by schema creation or cleanup scripts).

- Modernization File Inputs/Outputs: IN (DSN: AWS.M2.CARDDEMO.ESDSRRDS.PS, DISP: SHR), OUT (DSN: AWS.M2.CARDDEMO.USRSEC.VSAM.RRDS, DISP: SHR)

JCL Job: FTPJCLS

- **Job Stream Description:** Sequential execution of batch components. Map to Spring Batch tasklets or AWS Batch definition configurations.

- **Steps Defined (1):**

- **Step STEP1:** Runs FTP (Executes program FTP).

JCL Job: INTCALC

- **Job Stream Description:** Sequential execution of batch components. Map to Spring Batch tasklets or AWS Batch definition configurations.

- **Steps Defined (1):**

Step STEP15: Runs CBACT04C (Executes target application COBOL program CBACT04C).

- Modernization File Inputs/Outputs: STEPLIB (DSN: AWS.M2.CARDDDEM0.LOADLIB, DISP: SHR), TCATBALF (DSN: AWS.M2.CARDDDEM0.TCATBALF.VSAM.KSDS, DISP: SHR), XREFFILE (DSN: AWS.M2.CARDDDEM0.CARDXREF.VSAM.KSDS, DISP: SHR), XREFFIL1 (DSN: AWS.M2.CARDDDEM0.CARDXREF.VSAM.AIX.PATH, DISP: SHR), ACCTFILE (DSN: AWS.M2.CARDDDEM0.ACCTDATA.VSAM.KSDS, DISP: SHR), DISCGRP (DSN: AWS.M2.CARDDDEM0.DISCGRP.VSAM.KSDS, DISP: SHR), TRANSACT (DSN: AWS.M2.CARDDDEM0.SYSTRAN(, DISP: NEW, CATLG, DELETE)

JCL Job: INTRDRJ1

- **Job Stream Description:** Sequential execution of batch components. Map to Spring Batch tasklets or AWS Batch definition configurations.
- **Steps Defined (2):**

Step IDCAMS: Runs IDCAMS (VSAM Access Method Services utility; replaced by schema creation or cleanup scripts).

- Modernization File Inputs/Outputs: IN (DSN: AWS.M2.CARDEM0.FTP.TEST, DISP: SHR), OUT (DSN: AWS.M2.CARDEM0.FTP.TEST.BKUP, DISP: SHR)
- **Step STEP01:** Runs IEBGENER (System copy utility; replaced by standard OS file copy or cloud storage commands).

JCL Job: INTRDRJ2

- **Job Stream Description:** Sequential execution of batch components. Map to Spring Batch tasklets or AWS Batch definition configurations.
- **Steps Defined (1):**

Step IDCAMS: Runs IDCAMS (VSAM Access Method Services utility; replaced by schema creation or cleanup scripts).

- Modernization File Inputs/Outputs: IN (DSN: AWS.M2.CARDEM0.FTP.TEST.BKUP, DISP: SHR), OUT (DSN: AWS.M2.CARDEM0.FTP.TEST.BKUP.INTRDR, DISP: SHR)

JCL Job: OPENFIL

- **Job Stream Description:** Sequential execution of batch components. Map to Spring Batch tasklets or AWS Batch definition configurations.
- **Steps Defined (1):**
- **Step OPCIFIL:** Runs SDSF (Executes program SDSF).

JCL Job: POSTTRAN

- **Job Stream Description:** Sequential execution of batch components. Map to Spring Batch tasklets or AWS Batch definition configurations.
- **Steps Defined (1):**

Step STEP15: Runs CBTRN02C (Executes target application COBOL program CBTRN02C).

- Modernization File Inputs/Outputs: STEPLIB (DSN: AWS.M2.CARDDEMO.LOADLIB, DISP: SHR), TRANFILE (DSN: AWS.M2.CARDDEMO.TRANSACTION.VSAM.KSDS, DISP: SHR), DALYTRAN (DSN: AWS.M2.CARDDEMO.DALYTRAN.PS, DISP: SHR), XREFFILE (DSN: AWS.M2.CARDDEMO.CARDXREF.VSAM.KSDS, DISP: SHR), DALYREJS (DSN: AWS.M2.CARDDEMO.DALYREJS, DISP: NEW, CATLG, DELETE), ACCTFILE (DSN: AWS.M2.CARDDEMO.ACCTDATA.VSAM.KSDS, DISP: SHR), TCATBALF (DSN: AWS.M2.CARDDEMO.TCATBALF.VSAM.KSDS, DISP: SHR)

JCL Job: PRTCATBL

- **Job Stream Description:** Sequential execution of batch components. Map to Spring Batch tasklets or AWS Batch definition configurations.
- **Steps Defined (3):**

Step DELDEF: Runs IEFBR14 (System utility for file allocation/deletion; redundant in modernized environments).

- Modernization File Inputs/Outputs: THEFILE (DSN: AWS.M2.CARDDEMO.TCATBALF.REPT, DISP: MOD, DELETE)

- **Step STEP05R:** Runs REPROC (Executes program REPROC).

Step STEP10R: Runs SORT (Data sorting utility; replaced by DB order-by queries or local sorting logic).

- Modernization File Inputs/Outputs: SORTIN (DSN: AWS.M2.CARDDEMO.TCATBALF.BKUP, DISP: SHR), SORTOUT (DSN: AWS.M2.CARDDEMO.TCATBALF.REPT, DISP: NEW, CATLG, DELETE)

JCL Job: READACCT

- **Job Stream Description:** Sequential execution of batch components. Map to Spring Batch tasklets or AWS Batch definition configurations.
- **Steps Defined (2):**

Step PREDEL: Runs IEFBR14 (System utility for file allocation/deletion; redundant in modernized environments).

- Modernization File Inputs/Outputs: DD01 (DSN: AWS.M2.CARDDEMO.ACCTDATA.PSCOMP, DISP: MOD, DELETE, DELETE), DD02 (DSN: AWS.M2.CARDDEMO.ACCTDATA.ARRYPS, DISP: MOD, DELETE, DELETE), DD03 (DSN: AWS.M2.CARDDEMO.ACCTDATA.VBPS, DISP: MOD, DELETE, DELETE)

Step STEP05: Runs CBACT01C (Executes target application COBOL program CBACT01C).

- Modernization File Inputs/Outputs: STEPLIB (DSN: AWS.M2.CARDDemo.LOADLIB, DISP: SHR), ACCTFILE (DSN: AWS.M2.CARDDemo.ACCTDATA.VSAM.KSDS, DISP: SHR), OUTFILE (DSN: AWS.M2.CARDDemo.ACCTDATA.PSCOMP, DISP: NEW, CATLG, DELETE), ARRYFILE (DSN: AWS.M2.CARDDemo.ACCTDATA.ARRYPS, DISP: NEW, CATLG, DELETE), VBRCFILE (DSN: AWS.M2.CARDDemo.ACCTDATA.VBPS, DISP: NEW, CATLG, DELETE)

JCL Job: READCARD

- **Job Stream Description:** Sequential execution of batch components. Map to Spring Batch tasklets or AWS Batch definition configurations.
- **Steps Defined (1):**

Step STEP05: Runs CBACT02C (Executes target application COBOL program CBACT02C).

- Modernization File Inputs/Outputs: STEPLIB (DSN: AWS.M2.CARDDemo.LOADLIB, DISP: SHR), CARDFILE (DSN: AWS.M2.CARDDemo.CARDDATA.VSAM.KSDS, DISP: SHR)

JCL Job: READCUST

- **Job Stream Description:** Sequential execution of batch components. Map to Spring Batch tasklets or AWS Batch definition configurations.
- **Steps Defined (1):**

Step STEP05: Runs CBCUS01C (Executes target application COBOL program CBCUS01C).

- Modernization File Inputs/Outputs: STEPLIB (DSN: AWS.M2.CARDDemo.LOADLIB, DISP: SHR), CUSTFILE (DSN: AWS.M2.CARDDemo.CUSTDATA.VSAM.KSDS, DISP: SHR)

JCL Job: READXREF

- **Job Stream Description:** Sequential execution of batch components. Map to Spring Batch tasklets or AWS Batch definition configurations.
- **Steps Defined (1):**

Step STEP05: Runs CBACT03C (Executes target application COBOL program CBACT03C).

- Modernization File Inputs/Outputs: STEPLIB (DSN: AWS.M2.CARDDemo.LOADLIB, DISP: SHR), XREFFILE (DSN: AWS.M2.CARDDemo.CARDXREF.VSAM.KSDS, DISP: SHR)

JCL Job: REPTFILE

- **Job Stream Description:** Sequential execution of batch components. Map to Spring Batch tasklets or AWS Batch definition configurations.
- **Steps Defined (1):**

- **Step STEP05:** Runs IDCAMS (VSAM Access Method Services utility; replaced by schema creation or cleanup scripts).

JCL Job: TCATBALF

- **Job Stream Description:** Sequential execution of batch components. Map to Spring Batch tasklets or AWS Batch definition configurations.
- **Steps Defined (3):**
- **Step STEP05:** Runs IDCAMS (VSAM Access Method Services utility; replaced by schema creation or cleanup scripts).
- **Step STEP10:** Runs IDCAMS (VSAM Access Method Services utility; replaced by schema creation or cleanup scripts).

Step STEP15: Runs IDCAMS (VSAM Access Method Services utility; replaced by schema creation or cleanup scripts).

- Modernization File Inputs/Outputs: TCATBAL (DSN: AWS.M2.CARDDEM0.TCATBALF.PS, DISP: SHR), TCATBALV (DSN: AWS.M2.CARDDEM0.TCATBALF.VSAM.KSDS, DISP: SHR)

JCL Job: TRANBKP

- **Job Stream Description:** Sequential execution of batch components. Map to Spring Batch tasklets or AWS Batch definition configurations.
- **Steps Defined (3):**
- **Step STEP05R:** Runs REPROC (Executes program REPROC).
- **Step STEP05:** Runs IDCAMS (VSAM Access Method Services utility; replaced by schema creation or cleanup scripts).
- **Step STEP10:** Runs IDCAMS (VSAM Access Method Services utility; replaced by schema creation or cleanup scripts).

JCL Job: TRANCATG

- **Job Stream Description:** Sequential execution of batch components. Map to Spring Batch tasklets or AWS Batch definition configurations.
- **Steps Defined (3):**
- **Step STEP05:** Runs IDCAMS (VSAM Access Method Services utility; replaced by schema creation or cleanup scripts).
- **Step STEP10:** Runs IDCAMS (VSAM Access Method Services utility; replaced by schema creation or cleanup scripts).

Step STEP15: Runs IDCAMS (VSAM Access Method Services utility; replaced by schema creation or cleanup scripts).

- Modernization File Inputs/Outputs: TRANCATG (DSN: AWS.M2.CARDDemo.TRANCATG.PS, DISP: SHR), TCATVSAM (DSN: AWS.M2.CARDDemo.TRANCATG.VSAM.KSDS, DISP: SHR)

JCL Job: TRANFILE

- **Job Stream Description:** Sequential execution of batch components. Map to Spring Batch tasklets or AWS Batch definition configurations.
- **Steps Defined (8):**
- **Step CLCIFIL:** Runs SDSF (Executes program SDSF).
- **Step STEP05:** Runs IDCAMS (VSAM Access Method Services utility; replaced by schema creation or cleanup scripts).
- **Step STEP10:** Runs IDCAMS (VSAM Access Method Services utility; replaced by schema creation or cleanup scripts).
- **Step STEP15:** Runs IDCAMS (VSAM Access Method Services utility; replaced by schema creation or cleanup scripts).
 - Modernization File Inputs/Outputs: TRANSACT (DSN: AWS.M2.CARDDemo.DALYTRAN.PS.INIT, DISP: SHR), TRANVSAM (DSN: AWS.M2.CARDDemo.TRANSACT.VSAM.KSDS, DISP: SHR)
- **Step STEP20:** Runs IDCAMS (VSAM Access Method Services utility; replaced by schema creation or cleanup scripts).
- **Step STEP25:** Runs IDCAMS (VSAM Access Method Services utility; replaced by schema creation or cleanup scripts).
- **Step STEP30:** Runs IDCAMS (VSAM Access Method Services utility; replaced by schema creation or cleanup scripts).
- **Step OPCIFIL:** Runs SDSF (Executes program SDSF).

JCL Job: TRANIDX

- **Job Stream Description:** Sequential execution of batch components. Map to Spring Batch tasklets or AWS Batch definition configurations.
- **Steps Defined (3):**
- **Step STEP20:** Runs IDCAMS (VSAM Access Method Services utility; replaced by schema creation or cleanup scripts).
- **Step STEP25:** Runs IDCAMS (VSAM Access Method Services utility; replaced by schema creation or cleanup scripts).
- **Step STEP30:** Runs IDCAMS (VSAM Access Method Services utility; replaced by schema creation or cleanup scripts).

JCL Job: TRANREPT

- **Job Stream Description:** Sequential execution of batch components. Map to Spring Batch tasklets or AWS Batch definition configurations.
- **Steps Defined (3):**
- **Step STEP05R:** Runs REPROC (Executes program REPROC).

Step STEP05R: Runs SORT (Data sorting utility; replaced by DB order-by queries or local sorting logic).

- Modernization File Inputs/Outputs: SORTIN (DSN: AWS.M2.CARDDemo0.TRANSACT.BKUP(, DISP: SHR), SORTOUT (DSN: AWS.M2.CARDDemo0.TRANSACT.DALY(, DISP: NEW, CATLG, DELETE)

Step STEP10R: Runs CBTRN03C (Executes target application COBOL program CBTRN03C).

- Modernization File Inputs/Outputs: STEPLIB (DSN: AWS.M2.CARDDemo0.LOADLIB, DISP: SHR), TRANFILE (DSN: AWS.M2.CARDDemo0.TRANSACT.DALY(, DISP: SHR), CARDXREF (DSN: AWS.M2.CARDDemo0.CARDXREF.VSAM.KSDS, DISP: SHR), TRANTYPE (DSN: AWS.M2.CARDDemo0.TRANTYPE.VSAM.KSDS, DISP: SHR), TRANCATG (DSN: AWS.M2.CARDDemo0.TRANCATG.VSAM.KSDS, DISP: SHR), DATEPARM (DSN: AWS.M2.CARDDemo0.DATEPARM, DISP: SHR), TRANREPT (DSN: AWS.M2.CARDDemo0.TRANREPT(, DISP: NEW, CATLG, DELETE)

JCL Job: TRANTYPE

- **Job Stream Description:** Sequential execution of batch components. Map to Spring Batch tasklets or AWS Batch definition configurations.
- **Steps Defined (3):**
- **Step STEP05:** Runs IDCAMS (VSAM Access Method Services utility; replaced by schema creation or cleanup scripts).
- **Step STEP10:** Runs IDCAMS (VSAM Access Method Services utility; replaced by schema creation or cleanup scripts).

Step STEP15: Runs IDCAMS (VSAM Access Method Services utility; replaced by schema creation or cleanup scripts).

- Modernization File Inputs/Outputs: TRANTYPE (DSN: AWS.M2.CARDDemo0.TRANTYPE.PS, DISP: SHR), TTYPVSAM (DSN: AWS.M2.CARDDemo0.TRANTYPE.VSAM.KSDS, DISP: SHR)

JCL Job: TXT2PDF1

- **Job Stream Description:** Sequential execution of batch components. Map to Spring Batch tasklets or AWS Batch definition configurations.
- **Steps Defined (1):**

Step TXT2PDF: Runs IKJEFT1B (TSO monitor program executing DB2 batch commands; modernized as direct Spring/Java database tasks).

- Modernization File Inputs/Outputs: STEPLIB (DSN: AWS.M2.LBD.TXT2PDF.LOAD, DISP: SHR), SYSEXEC (DSN: AWS.M2.LBD.TXT2PDF.EXEC, DISP: SHR), INDD (DSN: AWS.M2.CARDDemo.STATEMNT.PS, DISP: SHR)

JCL Job: WAITSTEP

- **Job Stream Description:** Sequential execution of batch components. Map to Spring Batch tasklets or AWS Batch definition configurations.
- **Steps Defined (1):**

Step WAIT: Runs COBSWAIT (Executes target application COBOL program COBSWAIT).

- Modernization File Inputs/Outputs: STEPLIB (DSN: AWS.M2.CARDDemo.LOADLIB, DISP: SHR)

JCL Job: XREFFILE

- **Job Stream Description:** Sequential execution of batch components. Map to Spring Batch tasklets or AWS Batch definition configurations.
- **Steps Defined (6):**
- **Step STEP05:** Runs IDCAMS (VSAM Access Method Services utility; replaced by schema creation or cleanup scripts).
- **Step STEP10:** Runs IDCAMS (VSAM Access Method Services utility; replaced by schema creation or cleanup scripts).

Step STEP15: Runs IDCAMS (VSAM Access Method Services utility; replaced by schema creation or cleanup scripts).

- Modernization File Inputs/Outputs: XREFDATA (DSN: AWS.M2.CARDDemo.CARDXREF.PS, DISP: SHR), XREFVSAM (DSN: AWS.M2.CARDDemo.CARDXREF.VSAM.KSDS, DISP: SHR)
- **Step STEP20:** Runs IDCAMS (VSAM Access Method Services utility; replaced by schema creation or cleanup scripts).
- **Step STEP25:** Runs IDCAMS (VSAM Access Method Services utility; replaced by schema creation or cleanup scripts).
- **Step STEP30:** Runs IDCAMS (VSAM Access Method Services utility; replaced by schema creation or cleanup scripts).

4.6 Assembler Module Analysis

Lower-level HLASM modules referenced by subprograms in execution flows:

Assembler Module: COBDATFT

- **Description:** Assembler date formatting subroutine.

- **Entry Points:** COBDATFT
- **Calls Made:** None

Assembler Module: MVSWAIT

- **Description:** Assembler interval wait subroutine.
- **Entry Points:**
- **Calls Made:** None

4.7 CICS BMS Mapset Layouts

Defined CICS terminal screens, field layouts, and positioning mappings:

Mapset: COACTUP

- **Maps Included:** CACTUPA
- **Field Layouts (Top 20):** FILLER (Pos: N/A, Len: 0), TRNNAME (Pos: N/A, Len: 0), TITLE01 (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CURDATE (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), PGMNAME (Pos: N/A, Len: 0), TITLE02 (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CURTIME (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), ACCTSID (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), ACSTTUS (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), OPNYEAR (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 1)

Mapset: COACTVW

- **Maps Included:** CACTVWA
- **Field Layouts (Top 20):** FILLER (Pos: N/A, Len: 0), TRNNAME (Pos: N/A, Len: 0), TITLE01 (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CURDATE (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), PGMNAME (Pos: N/A, Len: 0), TITLE02 (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CURTIME (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), ACCTSID (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), ACSTTUS (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), ADTOPEN (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0)

Mapset: COADM01

- **Maps Included:** COADM1A
- **Field Layouts (Top 20):** FILLER (Pos: N/A, Len: 0), TRNNAME (Pos: N/A, Len: 0), TITLE01 (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CURDATE (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), PGMNAME (Pos: N/A, Len: 0), TITLE02 (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CURTIME (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), OPTN001 (Pos: N/A, Len: 0), OPTN002

(Pos: N/A, Len: 0), OPTN003 (Pos: N/A, Len: 0), OPTN004 (Pos: N/A, Len: 0), OPTN005 (Pos: N/A, Len: 0), OPTN006 (Pos: N/A, Len: 0), OPTN007 (Pos: N/A, Len: 0), OPTN008 (Pos: N/A, Len: 0), OPTN009 (Pos: N/A, Len: 0)

Mapset: COBIL00

- **Maps Included:** COBIL0A
- **Field Layouts (Top 20):** FILLER (Pos: N/A, Len: 0), TRNNAME (Pos: N/A, Len: 0), TITLE01 (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CURDATE (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), PGMNAME (Pos: N/A, Len: 0), TITLE02 (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CURTIME (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), ACTIDIN (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CURBAL (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CONFIRM (Pos: N/A, Len: 0)

Mapset: COCRDLI

- **Maps Included:** CCRDLIA
- **Field Layouts (Top 20):** FILLER (Pos: N/A, Len: 0), TRNNAME (Pos: N/A, Len: 0), TITLE01 (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CURDATE (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), PGMNAME (Pos: N/A, Len: 0), TITLE02 (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CURTIME (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 5), PAGENO (Pos: N/A, Len: 3), FILLER (Pos: N/A, Len: 0), ACCTSID (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CARDSID (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0)

Mapset: COCRDSL

- **Maps Included:** CCRDSLA
- **Field Layouts (Top 20):** FILLER (Pos: N/A, Len: 0), TRNNAME (Pos: N/A, Len: 0), TITLE01 (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CURDATE (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), PGMNAME (Pos: N/A, Len: 0), TITLE02 (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CURTIME (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), ACCTSID (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CARDSID (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CRDNAME (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0)

Mapset: COCRDUP

- **Maps Included:** CCRDUPA
- **Field Layouts (Top 20):** FILLER (Pos: N/A, Len: 0), TRNNAME (Pos: N/A, Len: 0), TITLE01 (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CURDATE (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), PGMNAME (Pos: N/A, Len: 0), TITLE02 (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0),

CURTIME (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), ACCTSID (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CARDSID (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CRDNAME (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0)

Mapset: COMEN01

- **Maps Included:** COMEN1A
- **Field Layouts (Top 20):** FILLER (Pos: N/A, Len: 0), TRNNAME (Pos: N/A, Len: 0), TITLE01 (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CURDATE (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), PGMNAME (Pos: N/A, Len: 0), TITLE02 (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CURTIME (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), OPTN001 (Pos: N/A, Len: 0), OPTN002 (Pos: N/A, Len: 0), OPTN003 (Pos: N/A, Len: 0), OPTN004 (Pos: N/A, Len: 0), OPTN005 (Pos: N/A, Len: 0), OPTN006 (Pos: N/A, Len: 0), OPTN007 (Pos: N/A, Len: 0), OPTN008 (Pos: N/A, Len: 0), OPTN009 (Pos: N/A, Len: 0)

Mapset: CORPT00

- **Maps Included:** CORPT0A
- **Field Layouts (Top 20):** FILLER (Pos: N/A, Len: 0), TRNNAME (Pos: N/A, Len: 0), TITLE01 (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CURDATE (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), PGMNAME (Pos: N/A, Len: 0), TITLE02 (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CURTIME (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), MONTHLY (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), YEARLY (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CUSTOM (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0)

Mapset: COSGN00

- **Maps Included:** COSGN0A
- **Field Layouts (Top 20):** FILLER (Pos: N/A, Len: 0), TRNNAME (Pos: N/A, Len: 0), TITLE01 (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CURDATE (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), PGMNAME (Pos: N/A, Len: 0), TITLE02 (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CURTIME (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), APPLID (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), SYSID (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0)

Mapset: COTRN00

- **Maps Included:** COTRN0A
- **Field Layouts (Top 20):** FILLER (Pos: N/A, Len: 0), TRNNAME (Pos: N/A, Len: 0), TITLE01 (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CURDATE (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0)

Len: 0), PGMNAME (Pos: N/A, Len: 0), TITLE02 (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CURTIME (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), PAGENUM (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), TRNIDIN (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0)

Mapset: COTRN01

- **Maps Included:** COTRN1A
- **Field Layouts (Top 20):** FILLER (Pos: N/A, Len: 0), TRNNAME (Pos: N/A, Len: 0), TITLE01 (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CURDATE (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), PGMNAME (Pos: N/A, Len: 0), TITLE02 (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CURTIME (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), TRNIDIN (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), TRNID (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CARDNUM (Pos: N/A, Len: 0)

Mapset: COTRN02

- **Maps Included:** COTRN2A
- **Field Layouts (Top 20):** FILLER (Pos: N/A, Len: 0), TRNNAME (Pos: N/A, Len: 0), TITLE01 (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CURDATE (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), PGMNAME (Pos: N/A, Len: 0), TITLE02 (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CURTIME (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), ACTIDIN (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CARDNIN (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0)

Mapset: COUSR00

- **Maps Included:** COUSR0A
- **Field Layouts (Top 20):** FILLER (Pos: N/A, Len: 0), TRNNAME (Pos: N/A, Len: 0), TITLE01 (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CURDATE (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), PGMNAME (Pos: N/A, Len: 0), TITLE02 (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CURTIME (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), PAGENUM (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), USRIDIN (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0)

Mapset: COUSR01

- **Maps Included:** COUSR1A

- **Field Layouts (Top 20):** FILLER (Pos: N/A, Len: 0), TRNNAME (Pos: N/A, Len: 0), TITLE01 (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CURDATE (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), PGMNAME (Pos: N/A, Len: 0), TITLE02 (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CURTIME (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FNAME (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), LNAME (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), USERID (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0)

Mapset: COUSR02

- **Maps Included:** COUSR2A
- **Field Layouts (Top 20):** FILLER (Pos: N/A, Len: 0), TRNNAME (Pos: N/A, Len: 0), TITLE01 (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CURDATE (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), PGMNAME (Pos: N/A, Len: 0), TITLE02 (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CURTIME (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), USRIDIN (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FNAME (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), LNAME (Pos: N/A, Len: 0)

Mapset: COUSR03

- **Maps Included:** COUSR3A
- **Field Layouts (Top 20):** FILLER (Pos: N/A, Len: 0), TRNNAME (Pos: N/A, Len: 0), TITLE01 (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CURDATE (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), PGMNAME (Pos: N/A, Len: 0), TITLE02 (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CURTIME (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), USRIDIN (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FNAME (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), LNAME (Pos: N/A, Len: 0)